

Problem Set # I

S14.3.2

Let $\vec{n} = \hat{x} \sin\theta \cos\phi + \hat{y} \sin\theta \sin\phi + \hat{z} \cos\theta$

i. $\vec{\sigma} \cdot \vec{n} = \begin{pmatrix} \cos\theta & \sin\theta (\cos\phi - i \sin\phi) \\ \sin\theta (\cos\phi + i \sin\phi) & -\cos\theta \end{pmatrix}$

$$= \begin{pmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{pmatrix}$$

$$\det(\vec{\sigma} \cdot \vec{n} - \lambda) = -(\cos\theta - \lambda)(\cos\theta + \lambda) - \sin^2\theta$$

$$= -1 + \lambda^2 = 0$$

The eigenvalues of $\vec{\sigma} \cdot \vec{n}$ are ± 1
(as expected)

$\lambda = 1$

$$\begin{pmatrix} \cos\theta - 1 & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta - 1 \end{pmatrix} |\vec{n}, +\rangle = 0$$

$$|\vec{n}, +\rangle = \begin{pmatrix} \sin\theta e^{i\phi} \\ 1 - \cos\theta \end{pmatrix} N$$

$$N^2 [\sin^2\theta + (1 - \cos\theta)^2] = 2N^2 (1 - \cos\theta)$$

$$= 4N^2 \sin^2\frac{\theta}{2}$$

$$N = \frac{1}{2 \sin\frac{\theta}{2}} e^{i\alpha}$$

α is an
arbitrary phase

$$|\vec{n}, +\rangle = \begin{pmatrix} \cos \theta/2 & e^{-i\varphi} \sin \theta/2 \\ \sin \theta/2 & e^{i\varphi/2} \end{pmatrix} e^{i\alpha}$$

$$= \begin{pmatrix} \cos \theta/2 & e^{-i\varphi/2} \sin \theta/2 \\ \sin \theta/2 & e^{i\varphi/2} \end{pmatrix} \quad \text{for } \alpha = \varphi/2$$

$$\lambda = -1 \quad \begin{pmatrix} \cos \theta + 1 & \sin \theta e^{-i\varphi} \\ \sin \theta e^{i\varphi} & -\cos \theta + 1 \end{pmatrix} |\vec{n}, -\rangle = 0$$

$$|\vec{n}, -\rangle = \begin{pmatrix} -\sin \theta e^{-i\varphi} \\ 1 + \cos \theta \end{pmatrix} N$$

$$= \begin{pmatrix} -\sin \theta/2 e^{-i\varphi} \\ \cos \theta/2 \end{pmatrix} e^{i\alpha'}$$

$$= \begin{pmatrix} -\sin \theta/2 e^{-i\varphi/2} \\ \cos \theta/2 e^{i\varphi/2} \end{pmatrix}$$

$$\text{for } \alpha' = \varphi/2$$

$$\langle \vec{n}, + | \vec{S} | \vec{n}, + \rangle$$

$$= \frac{\hbar}{2} (\cos \frac{\theta}{2} e^{i\varphi}, \sin \frac{\theta}{2}) \begin{bmatrix} (0 & 1) \\ (1 & 0) \end{bmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi} \\ \sin \frac{\theta}{2} \end{pmatrix}$$

$$= \frac{\hbar}{2} [\cos \frac{\theta}{2} \sin \frac{\theta}{2} 2 \cos \varphi, \cos \frac{\theta}{2} \sin \frac{\theta}{2} (-i) 2i \sin \varphi, \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}]$$

$$= \frac{\hbar}{2} \vec{n}$$

$$\langle \vec{n}, - | \vec{S} | \vec{n}, - \rangle$$

$$= \frac{\hbar}{2} (-\sin \frac{\theta}{2} e^{i\varphi}, \cos \frac{\theta}{2}) \begin{bmatrix} (0 & 1) \\ (1 & 0) \end{bmatrix} \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\varphi} \\ \cos \frac{\theta}{2} \end{pmatrix}$$

$$= \frac{\hbar}{2} [-\sin \frac{\theta}{2} \cos \frac{\theta}{2} 2 \cos \varphi, -\sin \frac{\theta}{2} \cos \frac{\theta}{2} (-i) 2i \sin \varphi, \sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}]$$

$$= -\frac{\hbar}{2} \vec{n}$$

14.3.6

1. A rotation by angle θ about $\hat{y}$ followed by a rotation of angle ϕ about (the old) $\hat{z}$, brings the direction $\hat{z}$ to the direction

$$\vec{m} = (\sin\theta \cos\phi, \sin\theta \sin\phi, \cos\theta)$$

Thus

$$U(\phi\hat{z})U(\theta\hat{y})S_zU(\theta\hat{y})U(\phi\hat{z}) = \vec{m} \cdot \vec{S}$$

We have

$$S_z |S_z = \hbar/2\rangle = \hbar/2 |S_z = \hbar/2\rangle$$

Therefore

$$\begin{aligned} \vec{m} \cdot \vec{S} U(\phi\hat{z})U(\theta\hat{y}) |S_z = \hbar/2\rangle \\ = \frac{\hbar}{2} U(\phi\hat{z})U(\theta\hat{y}) |S_z = \hbar/2\rangle \end{aligned}$$

Thus

$$U(\phi\hat{z})U(\theta\hat{y}) |S_z = \hbar/2\rangle = |m, +\rangle$$

$$U(\varphi \hat{z}) U(\theta \hat{y}) |S_z = \hbar/2\rangle$$

$$= e^{-i\varphi/2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} e^{-i\theta/2} \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\varphi/2} & 0 \\ 0 & e^{+i\varphi/2} \end{pmatrix} \begin{pmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} e^{-i\varphi/2} \cos\theta/2 \\ e^{+i\varphi/2} \sin\theta/2 \end{pmatrix} \quad \checkmark$$

S14.3.8

a. An arbitrary 2×2 matrix M can be written

$$M = m_0 + \vec{m} \cdot \vec{\sigma} = \begin{pmatrix} m_0 + m_3 & m_1 - i m_2 \\ m_1 + i m_2 & m_0 - m_3 \end{pmatrix}$$

where m_0, m_1, m_2 and m_3 are four complex numbers.

$$\begin{aligned} [M, \sigma_j] &= m_k [\sigma_k, \sigma_j] = 2i m_k \epsilon_{kjl} \sigma_l \\ &= 2i (\vec{\sigma} \times \vec{m})_j \end{aligned}$$

$$[M, \sigma_j] = 0 \quad (\text{for } j=1,2,3) \text{ requires } \vec{m} = 0$$

$$\text{Then } M = m_0 = \begin{pmatrix} m_0 & 0 \\ 0 & m_0 \end{pmatrix}$$

$$\begin{aligned} \text{b. } \{M, \sigma_j\} &= m_k \{\sigma_k, \sigma_j\} + 2m_0 \sigma_j \\ &= 2(m_j + m_0 \sigma_j) \end{aligned}$$

$$= 0 \quad \text{requires } m_0 = 0 \text{ and } m_j = 0$$

$$\{M, \sigma_j\} = 0 \quad (j=1,2,3) \text{ requires } M = 0$$

S14.4.3

$$\vec{B} = B(\cos \omega t \hat{x} + \sin \omega t \hat{y}) + B_0 \hat{z}$$

$$B \ll B_0$$

$$i\hbar \partial_t |\psi(t)\rangle = -\gamma \frac{\hbar}{2} \vec{\sigma} \cdot \vec{B}(t) |\psi(t)\rangle$$

$$\begin{aligned} \text{Let } |\Psi_r(t)\rangle &= e^{-i\omega t S_z / \hbar} |\psi(t)\rangle \\ &= e^{-i\omega t \sigma_z / 2} |\psi(t)\rangle \end{aligned}$$

$$i\hbar \partial_t |\Psi_r(t)\rangle = e^{-i\omega t \sigma_z / 2} \left[\frac{\omega \hbar \sigma_z}{2} - \gamma \frac{\hbar}{2} \vec{\sigma} \cdot \vec{B}(t) \right] |\psi(t)\rangle$$

$$\begin{aligned} &= \left[\frac{\hbar}{2} (\omega - \gamma B_0) \sigma_z - \gamma \frac{\hbar}{2} (\sigma_x \cos \omega t + \sigma_y \sin \omega t) B_x(t) \right. \\ &\quad \left. - \gamma \frac{\hbar}{2} (\sigma_y \cos \omega t - \sigma_x \sin \omega t) B_y(t) \right] e^{-i\omega t \sigma_z / 2} |\psi(t)\rangle \end{aligned}$$

$$\text{Since } e^{-i\theta \sigma_z / 2} \begin{pmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \end{pmatrix} e^{i\theta \sigma_z / 2} = \begin{pmatrix} \sigma_x \cos \theta + \sigma_y \sin \theta \\ \sigma_y \cos \theta - \sigma_x \sin \theta \\ \sigma_z \end{pmatrix}$$

($\theta = \omega t$ here)

$$B_x(t) = B \cos \omega t$$

$$B_y(t) = B \sin \omega t$$

$$i\hbar \partial_t |\Psi_r(t)\rangle = -\gamma \frac{\hbar}{2} \left[\left(B_0 - \frac{\omega}{\gamma}\right) \sigma_z + B \sigma_x \right] |\Psi_r(t)\rangle$$

time independent
hamiltonian H_I

$$|\Psi_r(t)\rangle = e^{-\frac{i}{\hbar} H_I t} |\Psi_r(0)\rangle$$

$$U_r(t) = e^{-\frac{i}{\hbar} H_I t} = e^{+\frac{i\gamma}{2} t \left[\left(B_0 - \frac{\omega}{\gamma}\right) \sigma_z + B \sigma_x \right]}$$

$$= \cos \frac{\theta}{2} + i(\vec{n} \cdot \vec{\sigma}) \sin \frac{\theta}{2}$$

$$\text{with } \theta \vec{n}' = \gamma t \left[\left(B_0 - \frac{\omega}{\gamma}\right) \hat{z} + B \hat{x} \right]$$

$$\theta = \gamma t \sqrt{\left(B_0 - \frac{\omega}{\gamma}\right)^2 + B^2} \equiv \omega_I t, \quad \omega_0 \equiv \gamma B_0$$

$$\vec{n}' = \frac{1}{\sqrt{\left(B_0 - \frac{\omega}{\gamma}\right)^2 + B^2}} \left[\left(B_0 - \frac{\omega}{\gamma}\right) \hat{z} + B \hat{x} \right]$$

$$= \frac{1}{\omega_I} \left[(\omega_0 - \omega) \hat{z} + \gamma B \hat{x} \right]$$

$$|\psi(t)\rangle = e^{i\omega t \sigma_z/2} U_r(t) \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\omega t/2} & 0 \\ 0 & e^{-i\omega t/2} \end{pmatrix} \begin{pmatrix} \cos(\frac{\omega_r t}{2}) + i n'_z \sin(\frac{\omega_r t}{2}) & i n'_x \sin(\frac{\omega_r t}{2}) \\ i n'_x \sin(\frac{\omega_r t}{2}) & \cos(\frac{\omega_r t}{2}) - i n'_z \sin(\frac{\omega_r t}{2}) \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\omega t/2} \left[\cos(\frac{\omega_r t}{2}) + i \frac{\omega_0 - \omega}{\omega_r} \sin(\frac{\omega_r t}{2}) \right] \\ e^{-i\omega t/2} i \frac{\gamma B}{\omega_r} \sin(\frac{\omega_r t}{2}) \end{pmatrix}$$

$$|\hat{n}, +\rangle = \begin{pmatrix} \cos \theta/2 & e^{-i\varphi/2} \\ \sin \theta/2 & e^{i\varphi/2} \end{pmatrix}$$

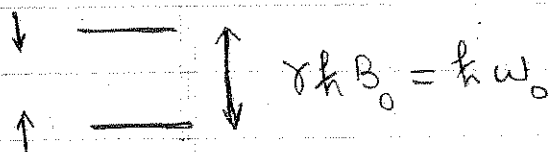
If $\omega = \omega_0$, the $|\psi(t)\rangle$ is

the eigenstate of spin $\vec{S} \cdot \vec{n} = +\frac{\hbar}{2}$ with $\vec{n}$

in the direction: $\theta = \omega_r t$ and $\varphi = \omega t$

$$\omega_r = \gamma B \ll \gamma B_0 = \omega$$

The spin flips sign with period $\frac{4\pi}{\omega_r}$.



The energy difference between the two levels is $\hbar \omega_0$.

When the frequency ω of the transverse component of $\vec{B}(t)$ equals ω_0 , there is resonant behaviour.

$$\begin{aligned}
 \langle \mu_z(t) \rangle &= -\gamma \frac{\hbar}{2} \langle \psi(t) | \sigma_z | \psi(t) \rangle \\
 &= -\gamma \frac{\hbar}{2} \left[\cos^2\left(\frac{\omega_r t}{2}\right) + \left(\frac{\omega_0 - \omega}{\omega_r}\right)^2 \sin^2\left(\frac{\omega_r t}{2}\right) \right. \\
 &\quad \left. - \left(\frac{\gamma B}{\omega_r}\right)^2 \sin^2\left(\frac{\omega_r t}{2}\right) \right] \\
 &= -\gamma \frac{\hbar}{2} \frac{1}{\omega_r^2} \left[(\omega_0 - \omega)^2 + (\gamma B)^2 \cos \omega_r t \right] \\
 &= + \langle \mu_z(0) \rangle \frac{(\omega_0 - \omega)^2 + (\gamma B)^2 \cos \omega_r t}{(\omega_0 - \omega)^2 + (\gamma B)^2}
 \end{aligned}$$

with $\langle \mu_z(0) \rangle = -\gamma \frac{\hbar}{2}$

S14.4.5

i In the non-rotating frame

$$\begin{aligned}\vec{V}_i(t + \Delta t) &= \vec{V}_i(t) + \Delta \vec{V} \\ &= \vec{V}_i(t) + \left. \frac{d\vec{V}}{dt} \right|_i \Delta t\end{aligned}$$

In the rotating frame, let us choose the axis such that

$$\vec{V}_r(t) = \vec{V}_i(t)$$

Then

$$\begin{aligned}\vec{V}_r(t + \Delta t) &= \vec{V}_i(t + \Delta t) - \Delta \vec{\theta} \times \vec{V}_i(t + \Delta t) \\ &= \vec{V}_i(t) + \left. \frac{d\vec{V}}{dt} \right|_i \Delta t - \vec{\omega} \times \vec{V}_i(t) \Delta t + O(\Delta t)^2\end{aligned}$$

$$= \vec{V}_r(t) + \left. \frac{d\vec{V}}{dt} \right|_r \Delta t$$

Therefore

$$\left. \frac{d\vec{V}}{dt} \right|_r = \left. \frac{d\vec{V}}{dt} \right|_i - \vec{\omega} \times \vec{V}$$

$$\text{ii.} \quad \left. \frac{d\vec{l}}{dt} \right|_r = \left. \frac{d\vec{l}}{dt} \right|_i - \vec{\omega} \times \vec{l}$$

$$\vec{T}_i = \left. \frac{d\vec{l}}{dt} \right|_i = \gamma (\vec{l} \times \vec{B}_i)$$

$$\vec{T}_r = \left. \frac{d\vec{l}}{dt} \right|_r = \left. \frac{d\vec{l}}{dt} \right|_i - \vec{\omega} \times \vec{l}$$

$$= \gamma (\vec{l} \times \vec{B})_i - \vec{\omega} \times \vec{l}$$

$$= \vec{l} \times (\gamma \vec{B}_i + \vec{\omega})$$

$$\therefore \vec{B}_r = \vec{B}_i + \frac{1}{\gamma} \vec{\omega}$$